> Question 1: Record all commands, pipeline configurations, screenshots, commit history, and observations for each experiment.
>
> Marks: 5 | Course outcome: CO1 | Environment: Ubuntu 24.04 LTS (WSL2), Git 2.43.0, Jenkins 2.580.2 LTS

This document compiles the complete evidence of Lab 1 as executed by Meenakshi Rakesh: every command run on Ubuntu 24.04 LTS (WSL2), the Jenkins pipeline configuration, the Git commit history, screenshots of each step, and the observations drawn from them. All artefacts are also maintained in the GitHub repository https://github.com/meenakshi16rp/Devops-lab-1-.

1. Environment and tool verification (Q4)

tool-versions.txt

```
meenakshi@ubuntu:~$ whoami
meenakshi
meenakshi@ubuntu:~$ lsb_release -d
Description:    Ubuntu 24.04.3 LTS
meenakshi@ubuntu:~$ git --version
git version 2.43.0
meenakshi@ubuntu:~$ java -version | head -1
openjdk version "21.0.12.1" 2026-08-18
meenakshi@ubuntu:~$ jenkins --version
2.580.2
meenakshi@ubuntu:~$ systemctl is-active jenkins
active
```

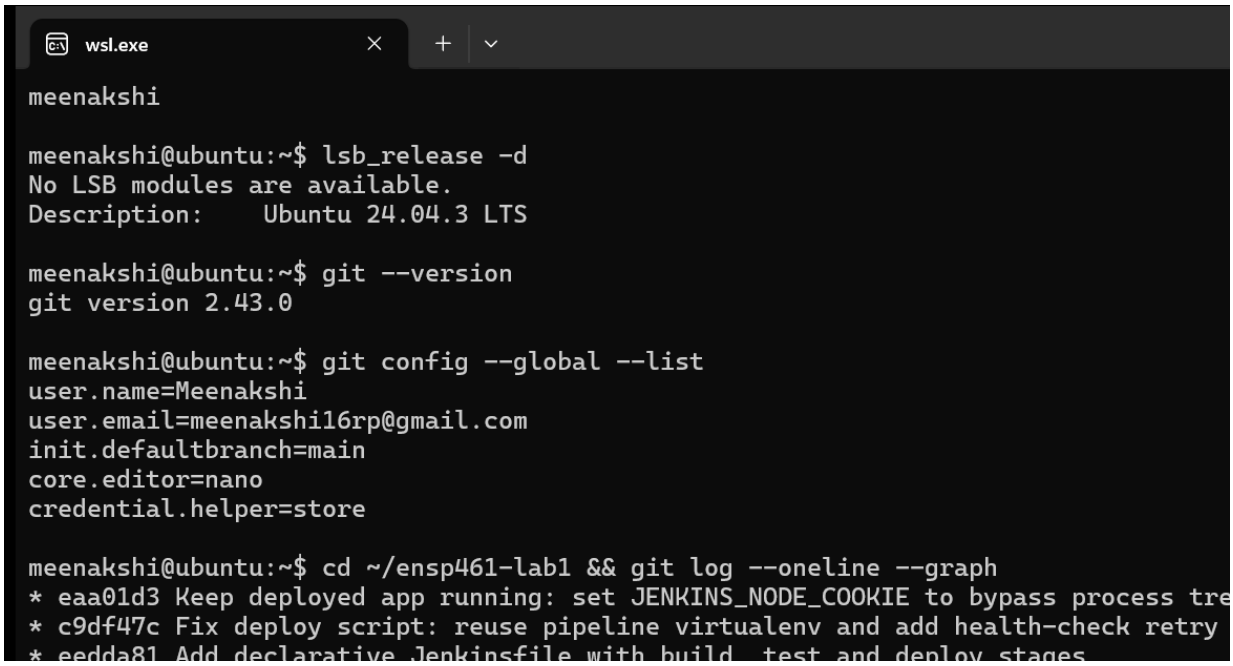


Figure: Terminal verification: user meenakshi, Ubuntu 24.04.3, git 2.43.0 and global Git configuration

2. Git installation and global configuration commands (Q4)

executed in WSL Ubuntu 24.04

```
sudo apt update && sudo apt install -y git
git config --global user.name "Meenakshi"
git config --global user.email "meenakshi16rp@gmail.com"
git config --global init.defaultBranch main
git config --global credential.helper store
git config --global --list
```

git-history-and-config.txt

```
=== git log --oneline --graph --all ===
* eaa01d3 Keep deployed app running: set JENKINS_NODE_COOKIE to bypass process tree killer
* c9df47c Fix deploy script: reuse pipeline virtualenv and add health-check retry loop
* eedda81 Add declarative Jenkinsfile with build, test and deploy stages
*   b4ead4e Merge feature/app-info-endpoint: add /info endpoint to main
|\
| * 946c34c Add /info endpoint with application metadata and unit test
|/
* 44c816b Add automated deployment script with health verification
* 6b3bf69 Initial commit: sample Flask application with unit tests

=== git config --global --list ===
user.name=Meenakshi
user.email=meenakshi16rp@gmail.com
init.defaultbranch=main
core.editor=nano
credential.helper=store

=== git remote -v ===
origin https://github.com/meenakshi16rp/Devops-lab-1-.git (fetch)
origin https://github.com/meenakshi16rp/Devops-lab-1-.git (push)

=== git branch -a ===
  feature/app-info-endpoint
* main
  remotes/origin/feature/app-info-endpoint
  remotes/origin/main
```

3. Repository commands - init, stage, commit, branch, merge, push (Q5)

commands executed under user meenakshi

```
cd ~/ensp461-lab1
git init -b main
git add README.md .gitignore app/app.py app/requirements.txt app/tests/test_app.py
git commit -m "Initial commit: sample Flask application with unit tests"
git add deploy/deploy.sh
git commit -m "Add automated deployment script with health verification"
git checkout -b feature/app-info-endpoint
# ... implemented /info endpoint + unit test on the branch ...
git add app/ && git commit -m "Add /info endpoint with application metadata and unit test"
git checkout main
git merge --no-ff feature/app-info-endpoint -m "Merge feature/app-info-endpoint: add /info \
      endpoint to main"
git add Jenkinsfile
git commit -m "Add declarative Jenkinsfile with build, test and deploy stages"
git remote add origin https://github.com/meenakshi16rp/Devops-lab-1-.git
git push -u origin main
git push -u origin feature/app-info-endpoint
```

resulting commit history (git log --graph)

```
=== git log --oneline --graph --all ===
* eaa01d3 Keep deployed app running: set JENKINS_NODE_COOKIE to bypass process tree killer
* c9df47c Fix deploy script: reuse pipeline virtualenv and add health-check retry loop
* eedda81 Add declarative Jenkinsfile with build, test and deploy stages
*   b4ead4e Merge feature/app-info-endpoint: add /info endpoint to main
|\
| * 946c34c Add /info endpoint with application metadata and unit test
|/
* 44c816b Add automated deployment script with health verification
* 6b3bf69 Initial commit: sample Flask application with unit tests

=== git config --global --list ===
user.name=Meenakshi
user.email=meenakshi16rp@gmail.com
init.defaultbranch=main
core.editor=nano
```

```
credential.helper=store

=== git remote -v ===
origin https://github.com/meenakshi16rp/Devops-lab-1-.git (fetch)
origin https://github.com/meenakshi16rp/Devops-lab-1-.git (push)

=== git branch -a ===
  feature/app-info-endpoint
* main
  remotes/origin/feature/app-info-endpoint
  remotes/origin/main
```

4. Jenkins pipeline configuration (Q6, Q7, Q3)

Jenkinsfile (committed to the repository)

```
// ENSP461 Lab 1 - Declarative CI/CD pipeline
// Stages: Build -> Test -> Deploy, triggered from GitHub
pipeline {
    agent any

    environment {
        VENV = "$WORKSPACE/.venv"
        DEPLOY_DIR = "$HOME/enp461-deploy"
    }

    stages {
        stage("Checkout") {
            steps {
                echo "Pulling source code from GitHub repository..."
                checkout scm
            }
        }

        stage("Build") {
            steps {
                echo "Building application: creating virtual environment and installing \
      dependencies..."
                sh """
                    python3 -m venv $VENV
                    $VENV/bin/pip install --quiet -r app/requirements.txt
                """
            }
        }

        stage("Test") {
            steps {
                echo "Running unit tests with pytest..."
                sh "$VENV/bin/python -m pytest app/tests -v"
            }
        }

        stage("Deploy") {
            steps {
                echo "Deploying application to local deployment directory..."
                sh "./deploy/deploy.sh"
            }
        }
    }

    post {
        success {
            echo "Pipeline SUCCESS: application built, tested and deployed."
        }
        failure {
            echo "Pipeline FAILED - check the stage logs."
        }
        always {
            echo "Cleaning workspace temporary files..."
            cleanWs()
        }
```

```
    }
}
```

job configuration summary

```
Jenkins job:        ensp461-lab1-pipeline (Pipeline)
Definition:         Pipeline script from SCM
SCM:                Git
Repository URL:     https://github.com/meenakshi16rp/Devops-lab-1-.git
Credentials:        github-creds (meenakshi16rp /****** GitHub PAT)
Branch:             */main
Script path:        Jenkinsfile
Build trigger:      Poll SCM -> H/5 * * * *
Post-build:         post { success / failure / always { cleanWs() } }
```

5. Build executions and console records (Q6, Q7, Q3)

build #1 console output (excerpt) - FAILURE

```
Started by user meenakshi
Obtained Jenkinsfile from git https://github.com/meenakshi16rp/Devops-lab-1-.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/ensp461-lab1-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-creds
Cloning the remote Git repository
Cloning repository https://github.com/meenakshi16rp/Devops-lab-1-.git
 > git init /var/lib/jenkins/workspace/ensp461-lab1-pipeline # timeout=10
Fetching upstream changes from https://github.com/meenakshi16rp/Devops-lab-1-.git
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials GitHub PAT for Meenakshi (ENSP461 Lab 1)
 > git fetch --tags --force --progress -- https://github.com/meenakshi16rp/Devops-lab-1-.git \
      +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git config remote.origin.url https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision eedda81de61a0d342439103a2e967508adde89ae (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f eedda81de61a0d342439103a2e967508adde89ae # timeout=10
Commit message: "Add declarative Jenki
```

build #2 console output (excerpt) - SUCCESS

```
Started by user meenakshi
Obtained Jenkinsfile from git https://github.com/meenakshi16rp/Devops-lab-1-.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/ensp461-lab1-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-creds
Cloning the remote Git repository
Cloning repository https://github.com/meenakshi16rp/Devops-lab-1-.git
 > git init /var/lib/jenkins/workspace/ensp461-lab1-pipeline # timeout=10
Fetching upstream changes from https://github.com/meenakshi16rp/Devops-lab-1-.git
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials GitHub PAT for Meenakshi (ENSP461 Lab 1)
 > git fetch --tags --force --progress -- https://github.com/meenakshi16rp/Devops-lab-1-.git \
      +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git config remote.origin.url https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision c9df47cad0b28b094c8b516b1c4648c5cbf80da2 (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f c9df47cad0b28b094c8b516b1c4648c5cbf80da2 # timeout=10
Commit message: "Fix deploy script: re
```

git-polling-log.txt - poll detects commit eaa01d3 -> build #3

```
Started on Nov 4, 2026, 6:14:33 PM
Using strategy: Default
[poll] Last Built Revision: Revision eaa01d3d856725c9504bf648a2f3704e02fe68b6 (refs/remotes/origin/main)
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-creds
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials GitHub PAT for Meenakshi (ENSP461 Lab 1)
 > git ls-remote -h -- https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
Found 2 remote heads on https://github.com/meenakshi16rp/Devops-lab-1-.git
[poll] Latest remote head revision on refs/heads/main is: eaa01d3d856725c9504bf648a2f3704e02fe68b6 - \
      already built by 3
Using strategy: Default
[poll] Last Built Revision: Revision eaa01d3d856725c9504bf648a2f3704e02fe68b6 (refs/remotes/origin/main)
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-creds
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials GitHub PAT for Meenakshi (ENSP461 Lab 1)
 > git ls-remote -h -- https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
Found 2 remote heads on https://github.com/meenakshi16rp/Devops-lab-1-.git
[poll] Latest remote head revision on refs/heads/main is: eaa01d3d856725c9504bf648a2f3704e02fe68b6 - \
      already built by 3
Done. Took 1.8 sec
No changes
```

6. Deployment verification commands (Q3)

deployed application verified after build #3

```
meenakshi@ubuntu:~$ curl -s http://localhost:5000/health
{"status":"UP"}
meenakshi@ubuntu:~$ curl -s http://localhost:5000/info
{"app":"ensp461-lab1-sample-app","author":"Meenakshi","pipeline":"Jenkins declarative \
      CI/CD","version":"1.0.0"}
```

7. Observations

- Build #1 failed in the Deploy stage (curl exit code 7): the deployment script invoked the system python3, which does not have Flask installed. Diagnosis from the console log and the deploy.log showed ModuleNotFoundError: No module named 'flask'.
- Fix: deploy.sh was changed to reuse the virtual environment created in the Build stage (VENV environment variable) and a health-check retry loop (10 attempts) was added. Committed as c9df47c; build #2 then succeeded end to end.
- The deployed server was killed when the build finished because Jenkins' process-tree killer reaps child processes. Setting JENKINS_NODE_COOKIE=dontKillMe before nohup keeps the application running after the build (commit eaa01d3).
- Build #3 was triggered automatically by the Poll SCM schedule - its cause reads Started by an SCM change - proving the automation works without manual intervention.
- Meaningful commit messages were used for every change so the history itself documents the work; the pipeline is fully reproducible from the repository (Jenkinsfile + deploy script).

8. Key screenshots (also embedded in the per-question reports)

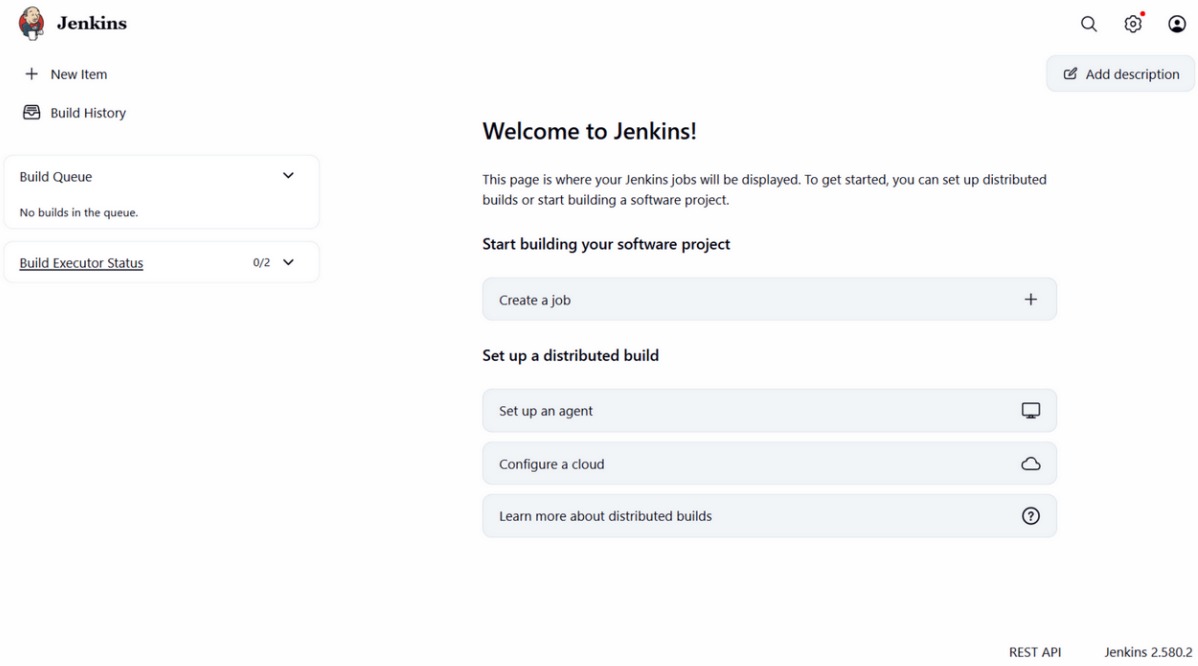


Figure: Jenkins 2.580.2 dashboard after setup, admin user meenakshi

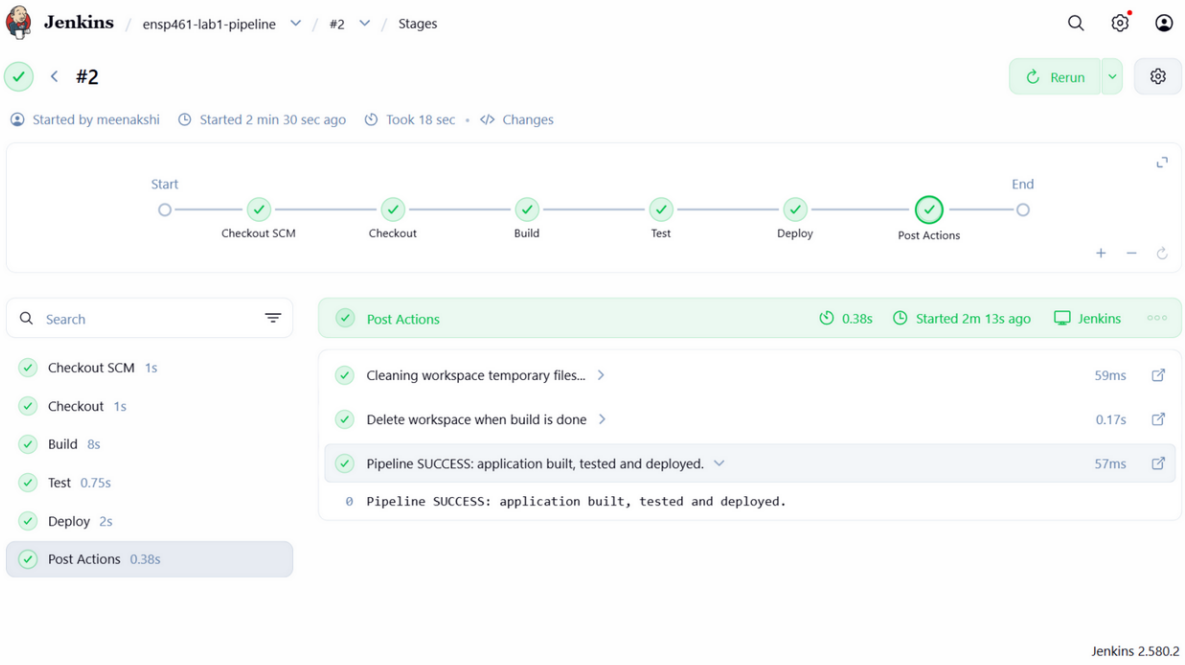


Figure: Build #2 stage graph: Checkout SCM, Checkout, Build, Test, Deploy, Post Actions - all green

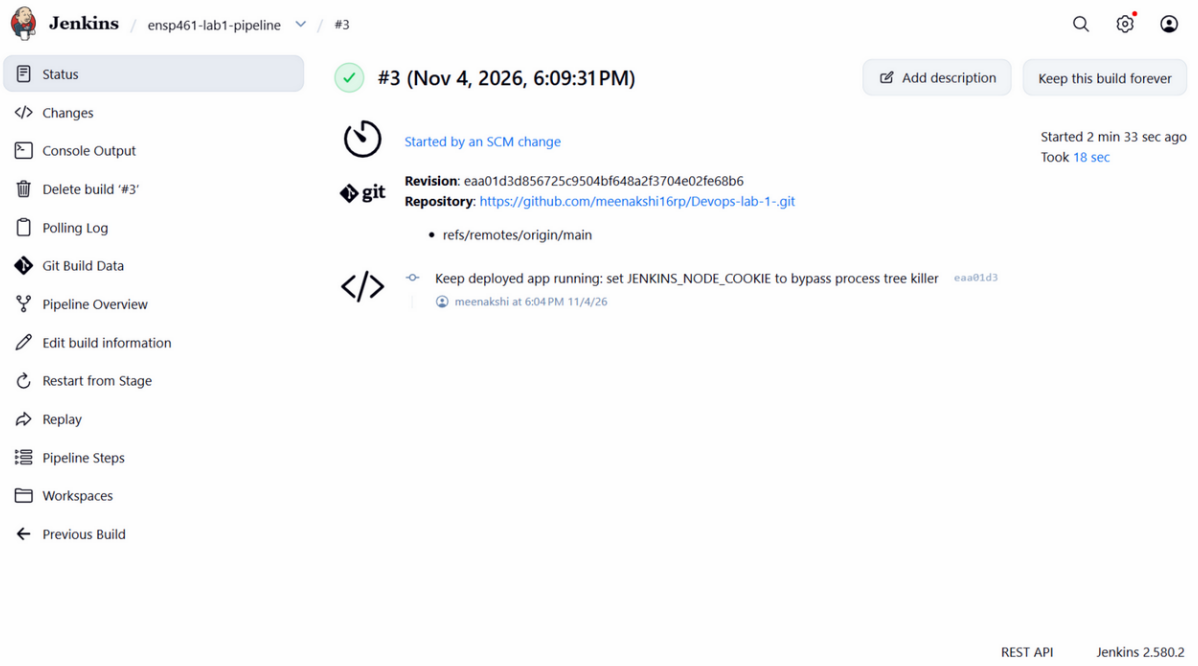


Figure: Build #3 automatically started by an SCM change (poll trigger)

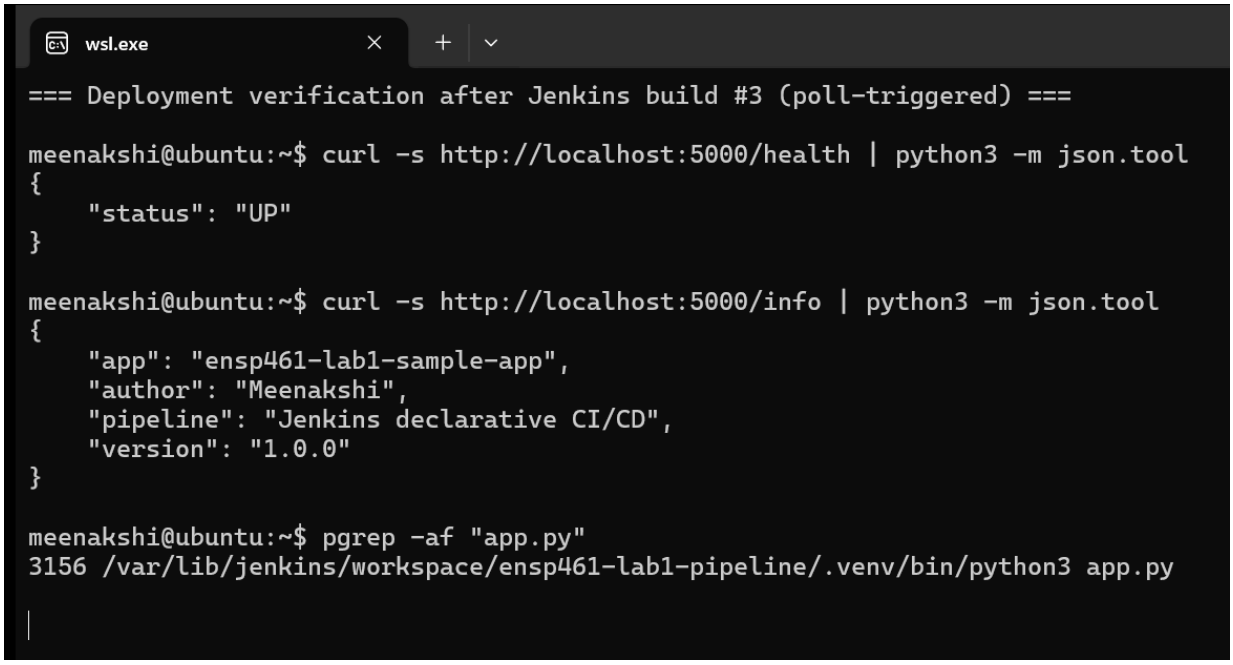


Figure: Deployed Flask application verified on port 5000 with author Meenakshi